

Cross-Platform Migration Blueprint

This reference document outlines the comprehensive procedural walkthrough required to transition the **Evidentia** local ecosystem seamlessly from a Linux (Ubuntu/Debian) native setup over to a Windows operating environment.

1. Windows Prerequisite Installation

Before moving code repositories, install the identical architectural runtimes matching your target windows machine:

- **Node.js (LTS Version):** Download and install from the official distribution portal. This registers system-wide variables for `npm`.
- **PHP (Thread Safe Zip):** Download from windows.php.net. Extract to an accessible route (e.g., `C:\\php`) and explicitly append this directory path inside the Windows **System Environment Variables (PATH)**.
- **Database Engine Stack:** Install **XAMPP** or **WampServer**. This automatically establishes localized background loops for Apache, MySQL, and phpMyAdmin out of the box.

2. Project Folder Packaging & Transfer

To avoid massive directory payloads and binary platform mismatches within the frontend workspace, follow this compression protocol:

CRITICAL RULE: Drop node_modules Before Zipping

The `node_modules/` folder inside your `frontend/` repository contains thousands of files customized uniquely to compile onto Linux architecture. Do NOT bundle this folder into your zip file. Your dependencies remain safely indexed inside your `package.json` configuration file.

1. Inside the Linux terminal, remove the cache tree:

```
rm -rf /home/kcplibrary/Documents/evidentia/frontend/node_modules
```

2. Zip the remaining root `evidentia/` directory.
3. Transfer the compressed archive onto the Windows drive using a local volume, cloud drive, or a Git remote stream.
4. Extract the directory anywhere on the Windows machine (e.g., `C:\\Users\\Username\\Documents\\evidentia`).

3. Database Structuring & Records Import

Migrate the schemas and testing credential rows (e.g., `admin` , `officer` entries):

1. On the source environment, visit the phpMyAdmin page at `http://localhost:8080/` .
2. Select `evidentia_db` , navigate to the **Export** pane, leave settings as Quick/SQL, and click Go to download the `.sql` script.
3. On Windows, start the XAMPP Control Panel and click 'Start' next to the MySQL option.
4. Navigate to the Windows phpMyAdmin dashboard, define a clean, empty database named `evidentia_db` , click **Import**, point it to your saved SQL dump, and hit run.

4. Initializing the Restored Ecosystem

Open Windows PowerShell or Command Prompt, and map out your optimized 3-terminal workstation sequence:

Workspace Terminal 1: React Frontend (Vite)

Navigate to your extracted frontend directory and reconstruct the windows package tree:

```
cd C:\\Users\\Username\\Documents\\evidentia\\frontend
npm install
npm run dev
```

Note: `npm install` reads `package.json` and compiles fresh Windows-native binaries. Vite will go live on `http://localhost:5173` .

Workspace Terminal 2: PHP API Handler

Spin up your internal PHP standalone gateway server inside your dedicated endpoint location:

```
cd C:\\Users\\Username\\Documents\\evidentia\\backend\\api
php -S localhost:8081
```

Note: This matches your optimized Linux architecture, hosting the live endpoints cleanly at `http://localhost:8081/login.php` .

Workspace Terminal 3: Persistent Database Access

If utilizing standard XAMPP or WampServer distributions, Apache and MySQL run directly as quiet Windows context actions. No active persistent terminal layout needs to stay open. Open any local web browser and log straight into your panels via: `http://localhost/phpmyadmin` .

5. Network Architecture & Target Ports Table

Your production ports line up cleanly without overlapping any platform configurations:

Service Name	Target URL / Port	Underlying Host Engine
React UI Gateway	<code>http://localhost:5173</code>	Vite Node Ecosystem
Authentication API	<code>http://localhost:8081</code>	PHP Standalone Server
phpMyAdmin GUI	<code>http://localhost/phpmyadmin</code>	XAMPP / Apache Engine
Relational Database	<code>localhost:3306</code>	MySQL Server Process